Hints: Linear algebra – 1. Determinants and invertibility

Homework

The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.

Problem 1

Hint 1. $B = A - J$, where J is the all-ones matrix, which has rank 1. Show that $\operatorname{rank} B^3 < 10$ (or $\operatorname{rank} B < 10$).

Hint 2. Expand $B^3 = (A-J)^3$. The term A^3 vanishes, and each of the remaining 7 terms contains a factor J, so has rank at most 1. Hence $\operatorname{rank} B^3 \leqslant 7 < 10$, so $\det B^3 = 0$.

Problem 2

Hint 1. Expand $(A+B)(A^{-1}+B^{-1}) = E$: you get $AB^{-1} + BA^{-1} = -E$. Put $X = AB^{-1}$; then $BA^{-1} = X^{-1}$.

Hint 2. $X + X^{-1} = -E$ means $X^2 + X + E = O$, hence $X^3 - E = (X-E)(X^2+X+E) = O$. So $(\det X)^3 = 1$, and over $\mathbb{R}$ this gives $\det X = 1$, i.e. $\det A = \det B$.

Hint 3. Over $\mathbb{C}$ the determinant of X can be a non-real cube root of unity. For 1×1 blocks, $\alpha^{-1} + 1 = (\alpha+1)^{-1}$ iff $\alpha^2 + \alpha + 1 = 0$. So with $\omega = e^{2\pi i/3}$, $B = E$ and A diagonal with entries in $\{\omega, \omega^2\}$ satisfy the relation; take $A = \omega E$ if $3 \nmid n$ and $A = \operatorname{diag}(\omega, \dots, \omega, \omega^2)$ if $3 \mid n$. Then $\det A \neq 1 = \det B$: the answer is no, for every $n \geqslant 2$.

Problem 3

Hint 1. Write $M = \begin{pmatrix} x & y \\ y & z \end{pmatrix}$. Then $a^2 - 4b = (x+z)^2 - 4(xz - y^2) = (x-z)^2 + (2y)^2$.

Hint 2. Since $x + z = a$ is fixed, M is determined by the point $(x - z, 2y)$, and the admissible points form the circle of radius $\sqrt{a^2 - 4b}$. So there is no M if $a^2 < 4b$, infinitely many if $a^2 > 4b$, and exactly one ($M = \frac{a}{2}E$) if $a^2 = 4b$. Answer: exactly the pairs with $a^2 = 4b$.

Problem 4

Hint 1. 2) $\Rightarrow$ 1) is immediate (take $B_i = A_i$). For the converse, use that A_i is symmetric: $\langle (\sum A_i^2)x, x\rangle = \sum \|A_i x\|^2$.

Hint 2. Take $x \neq 0$ in the kernel of $\sum A_i^2$; then $A_i x = 0$ for every i. So x is in the kernel of $\sum B_i^T A_i = \left(\sum A_i B_i\right)^T$ for any B_i.

Problem 5

Hint 1. $S = \sum A_i^T A_i$ is symmetric and $\langle Sx, x\rangle = \sum \|A_i x\|^2 \geqslant 0$. What does this say about its eigenvalues?

Hint 2. S is real symmetric, so its eigenvalues are real; if $Sx = \lambda x$ with $x \neq 0$, then $\lambda \|x\|^2 = \langle Sx, x\rangle \geqslant 0$, so $\lambda \geqslant 0$. Hence $\det S$, the product of the eigenvalues, is $\geqslant 0$.

Problem 6

Hint 1. A is singular, so $\operatorname{rank} \operatorname{adj} A \leqslant 1$.

Hint 2. For a matrix C of rank $\leqslant 1$, write $C = uv^T$ and use the rank-one update formula: $\det(E + C) = 1 + v^T u = 1 + \operatorname{tr} C$. Apply it to $C = \operatorname{adj} A$.

Problem 7

Hint 1. One direction is a computation: if $A = B^{-1}\bar{B}$ then $\bar{A} = \bar{B}^{-1}B = A^{-1}$. For the converse you need B invertible with $BA = \bar{B}$.

Hint 2. Try $B = zE + \bar{z}\,\bar{A}$ with $z \in \mathbb{C} \setminus \{0\}$. Using $\bar{A}A = E$, check that $BA = \bar{B}$.

Hint 3. $B = \bar{z}\,(\bar{A} + \frac{z}{\bar{z}}E)$, and $z/\bar{z}$ runs over the whole unit circle, while $\bar{A}$ has only finitely many eigenvalues. Choose z so that $-z/\bar{z}$ is not an eigenvalue.

Problem 8

Hint 1. Since $\det(AB) = \det A \det B$, it suffices to show that 0 is not an eigenvalue of AB. Let S be the set of eigenvalues of AB, and recall that AB and BA have the same characteristic polynomial, hence the same eigenvalues.

Hint 2. The eigenvalues of $BA = aE + bAB$ are the numbers $a + b\mu$, $\mu \in S$. So $f(\mu) = a + b\mu$ maps the finite set S onto itself, i.e. permutes it. Hence every $\mu \in S$ satisfies $f^k(\mu) = \mu$ for some $1 \leqslant k \leqslant |S| \leqslant n$.

Hint 3. $f^k(\mu) = b^k\mu + a(1 + b + \cdots + b^{k-1})$. If $0 \in S$, then $f^k(0) = 0$ gives $a(1 + b + \cdots + b^{k-1}) = 0$. But $a \neq 0$, and since $b \neq 1$ and $b^k \neq 1$, $1 + b + \cdots + b^{k-1} = \frac{b^k - 1}{b - 1} \neq 0$, a contradiction. So AB is invertible, and so are A and B. (The same computation shows that in fact every eigenvalue of AB equals $\frac{a}{1-b}$.)

Problem 9

Hint 1. The minors of order $n - 1$ are, up to sign, the entries of $\operatorname{adj} A$. Suppose $\det A = 0$; then $\operatorname{rank} \operatorname{adj} A \leqslant 1$.

Hint 2. If $\operatorname{adj} A = O$, all n^2 minors vanish. Otherwise $\operatorname{adj} A = uv^T$ with $u, v \neq 0$. If u has p zero coordinates and v has q, count the zero entries of uv^T: $np + nq - pq$. Since $k \geqslant 1$, $p + q \geqslant 1$, and this count is $\geqslant n > k$.

Problem 10

Hint 1. A non-zero (real) X with $AX = X(-A)$ exists if and only if A and $-A$ have a common eigenvalue (Sylvester's theorem on $AX = XB$; a complex solution gives a real one by taking real or imaginary parts). Since A is invertible, this means A has eigenvalues λ and $-\lambda$ with $\lambda \neq 0$.

Hint 2. $\operatorname{tr} \operatorname{adj} A = e_3(\lambda_1, \ldots, \lambda_4)$, the third elementary symmetric function of the eigenvalues. $A^2 + E$ is singular if and only if $\pm i$ are eigenvalues, which is such a pair.

Hint 3. Conversely, if the eigenvalues are $\lambda, -\lambda, \mu, \nu$, compute $e_1 = \mu + \nu$ and $e_3 = -\lambda^2(\mu + \nu)$. The condition $e_1 = e_3 \neq 0$ forces $\lambda^2 = -1$.

Problem 11

Hint 1. For odd n: $\det A = \det A^T = \det(-A) = (-1)^n \det A$, so $\det A = 0$.

Hint 2. For even n the determinant need not vanish (try $n = 2$). The non-zero eigenvalues of a real antisymmetric matrix are purely imaginary and come in pairs $\pm i\lambda$, so $\det A = \prod \lambda_k^2 \geqslant 0$. In fact $\det A$ is the square of a polynomial in the entries, the Pfaffian.

Problem 12

Hint 1. Use Hadamard's inequality: $|\det M| \leqslant \prod_j \|m_j\|$, where m_j are the columns of M. (Proof: for invertible M, Gram–Schmidt gives $M = QR$ with Q orthogonal and R upper triangular; then $|\det M| = \prod |r_{jj}|$ and $\|m_j\|^2 = \sum_{i \leqslant j} r_{ij}^2 \geqslant r_{jj}^2$.) Apply it to $M = E + A$.

Hint 2. The j-th column of $E + A$ has $\|m_j\|^2 = (1 + a_{jj})^2 + \sum_{i \neq j} a_{ij}^2 = 1 + x_j$ with $x_j = 2a_{jj} + \sum_i a_{ij}^2$. By $1 + x \leqslant e^x$, $\|m_j\| \leqslant e^{x_j/2}$, and multiplying over j gives $|\det(E + A)| \leqslant \exp\bigl(\sum_j a_{jj} + \frac{1}{2}\sum_{i,j} a_{ij}^2\bigr) = \exp\bigl(\operatorname{tr} A + \frac{1}{2}\sum_{i,j} a_{ij}^2\bigr)$.

Problem 13

Hint 1. Subtract the first row from each of the others. What can you say about the entries of rows $2, \ldots, n$ now?

Hint 2. They are even, so a factor 2 comes out of each of $n - 1$ rows. For sharpness, take the first row all ones and row i ($i \geqslant 2$) equal to all ones plus $2e_i$. After the subtraction its determinant is exactly 2^{n-1}.

Problem 14

Hint 1. Barbara wins. Look for a pairing strategy that forces two equal columns at the end.

Hint 2. Pair column j with column $j + 1004$. Whenever Alan writes x in position (i, j), Barbara writes the same x in the paired position in the same row. Check that the paired cell is always empty.

Problem 15

Hint 1. Player 0 wins. Player 0 places only four zeros, so aim for small zero patterns that force $\det = 0$ on their own: a zero row, a zero column, or a 2×2 block of zeros (rows i, j and columns k, l). Permuting rows, permuting columns and transposing preserve these patterns and $|\det|$, and they can move any cell to any other; so assume Player 1 opens at $(1, 1)$.

Hint 2. Player 0 answers at $(2, 2)$. After Player 1's second move at least one of $(2, 3)$, $(3, 2)$ is vacant; by transposing, say $(2, 3)$, and Player 0 plays there. Now Player 1 must take $(2, 1)$, otherwise Player 0 plays there and row 2 is zero.

Hint 3. Player 1 now holds $(1, 1)$, $(2, 1)$ and one more cell c. Look at the square of cells in rows $1, 3$ and columns $2, 3$: two zeros of it in one column complete a zero column (with row 2), and two zeros of it in one row complete a zero 2×2 block with $(2, 2), (2, 3)$. Player 0 puts the third zero at a cell z of this square, diagonally opposite to c if c lies in the square. Then the two neighbours of z in its row and column of the square are vacant: a double threat, and the fourth zero completes a pattern.

Problem 16

Hint 1. Changing entry (i, j) by t changes the determinant by t times the cofactor C_{ij}, since det is linear in each entry. So you need $\det A = 0$ with every cofactor non-zero, i.e. every entry of $\operatorname{adj} A$ non-zero.

Hint 2. Yes, for every n. If $\operatorname{rank} A = n - 1$, then $\operatorname{adj} A = uv^T$ with $Au = 0$ and $v^T A = 0$. Pick A whose kernel and left kernel are both spanned by $(1, \ldots, 1)$, e.g. $A = nE - J$; then $\operatorname{adj} A$ is a non-zero multiple of J. ($n = 1$ is trivial.)